

Lesson 7 Installation ROS in Docker

Note: Our Raspberry Pi image is already installed with ROS1 or ROS2 environment when you receive the package. This section is provided for learning and reference only.

1. ROS1 Installation

1.1 “melodic” Version (Need Networking)

1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker pull ros:melodic” to download the ROS1 image. The downloading takes time, please wait for a while with the patient.

```
pi@raspberrypi:~ $ docker pull ros:melodic
```

2) After the image downloading is complete, enter “docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name melodic -e DISPLAY=\${DISPLAY} --restart=always ros:melodic /bin/bash” in the terminal to run the container and name it “melodic”.

```
pi@raspberrypi:~ $ docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name melodic -e DISPLAY=${DISPLAY} --restart=always ros:melodic /bin/bash
75b6634e622762255058569dd769ed3925f77b9976539e85d1d70cd029929f9d
pi@raspberrypi:~ $ docker ps
```

3) Enter “xhost +” in the terminal to start the accessing control of “X Server”.

```
pi@raspberrypi:~ $ xhost +
access control disabled, clients can connect from any host
```

4) Enter “docker ps -a” in the terminal to check the ID of the newly created container.

```
pi@raspberrypi:~ $ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
75b6634e6227	ros:melodic	"/ros_entrypoint.sh ..."	15 seconds ago	Up 13 seconds		melodic
a24dbc912220	debian	"bash"	About an hour ago	Exited (0) About an hour ago		blissful_
mccarthy	debian:latest	"/bin/bash"	2 hours ago	Exited (255) 26 minutes ago		jolly_bas
f1b75532336	debian:latest	"/bin/bash"	2 hours ago	Exited (0) 2 hours ago		fervent_m
3313afaf6374	debian	"/bin/bash"	3 hours ago	Exited (255) 26 minutes ago		crazy_bel
32d0e43b41eb	debian:latest	"/bin/bash"	4 hours ago	Exited (255) 26 minutes ago		jolly_edi
9cfc6a98a68c	debian:latest	"/bin/bash"				son

5) Enter “docker exec -it 75b6634e6227 /bin/bash” in the terminal to access the container.

```
pi@raspberrypi:~$ docker exec -it 75b6634e6227 /bin/bash
root@raspberrypi:/#
```

6) Enter “useradd -m -s /bin/bash ubuntu” in the terminal to create a new user.

```
root@raspberrypi:/# useradd -m -s /bin/bash ubuntu
root@raspberrypi:/# useradd -m -s /bin/bash ubuntu
```

7) Enter “passwd ubuntu” in the terminal to set a password for “ubuntu”. After successfully set, the new password needs to be input twice.

```
root@50e095a38fff:/# passwd ubuntu
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
```

8) Enter “usermod -aG sudo ubuntu” in the terminal. Add “sudo” to the new user “ubuntu” to give it superuser privileges.

```
root@raspberrypi:/# usermod -aG sudo ubuntu
root@raspberrypi:/# usermod -aG sudo ubuntu
```

9) Enter “sudo apt-get update -y && sudo apt-get upgrade -y” in the terminal to update the available software package list and upgrade the installed package on the system.

```
root@raspberrypi:/# sudo apt-get update -y && sudo apt-get upgrade -y
Get:1 http://snapshots.ros.org/melodic/final/ubuntu bionic InRelease [13.8 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports bionic InRelease [242 kB]
Get:3 http://snapshots.ros.org/melodic/final/ubuntu bionic/main arm64 Packages [1017 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports bionic-updates InRelease [88.7 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports bionic-backports InRelease [83.3 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports bionic-security InRelease [88.7 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports bionic/universe arm64 Packages [11.0 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports bionic/main arm64 Packages [1285 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports bionic/restricted arm64 Packages [572 B]
```

10) Enter “sudo apt-get install vim -y” in the terminal to install the “Vim” text editor.

```
root@raspberrypi:/# sudo apt-get install vim -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libgpm2 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
```

11) Enter “sudo apt-get install ros-melodic-desktop-full -y” in the terminal to install the complete desktop environment of the “ROS Melodic” version. The downloading takes time, please wait for a while with the patient.

```
root@raspberrypi:/# sudo apt-get install ros-melodic-desktop-full -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  autoconf automake autopoint autotools-dev avahi-daemon bind9-host blt bsdmainutils curl cython dbus dh-autoreconf dh-strip-nondeterminism dmsetup file fltk1.3-doc fluid fonts-lato fonts-liberation font
```

1.2 ROS1 Environment Test

1) Enter the “docker exec -it -u ubuntu -w /home/ubuntu 75b6 /bin/bash” command to access the container. (Note: “75b6” is the ID of the container with ROS1 environment.)

```
pi@raspberrypi:~$ docker exec -it -u ubuntu -w /home/ubuntu 75b6 /bin/bash
```

2) Enter the “source /opt/ros/melodic/setup.bash” command to manually configure the ROS1 environment.

```
ubuntu@raspberrypi:~$ source /opt/ros/melodic/setup.bash
ubuntu@raspberrypi:~$
```

3) Every time you open the terminal, execute step 2 to load the working space. Enter the “echo “source /opt/ros/melodic/setup.bash” >> ~/.bashrc” command and write it into the “.bashrc” file.

```
ubuntu@raspberrypi:~$ echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
ubuntu@raspberrypi:~$
```

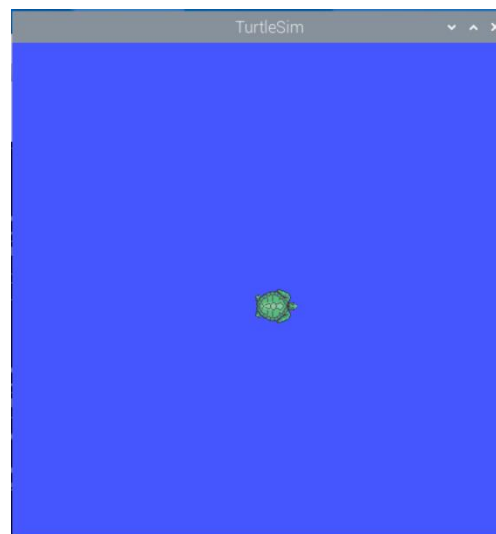
4) Then enter the “source ~/.bashrc” command to take the “.bashrc” file into effect. This action frees you from loading the working space environment

every time opening the terminal.

```
ubuntu@raspberrypi:~$ source ~/.bashrc
ubuntu@raspberrypi:~$
```

5) Enter “roslaunch turtlesim turtlesim_node” to start the turtle GUI interface. Its successful starting indicates the successful installation of ROS1.

```
ubuntu@raspberrypi:~$ roslaunch turtlesim turtlesim_node
```



2. ROS2 Installation

2.1 “humble” Version (Need Networking)

1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker pull ros:humble” to download the ROS2 image. The downloading takes time, please wait for a while with patient.

```
pi@raspberrypi:~ $ docker pull ros:humble
```

2) After the image downloading is complete, enter “docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name humble -e DISPLAY=\${DISPLAY} --restart=always ros:humble /bin/bash” in the terminal to run the container and name it “humble”.

```
pi@raspberrypi:~$ docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name melodic -e DISPLAY=${DISPLAY} --restart=always ros:melodic /bin/bash
75b6634e622762255058569dd769ed3925f77b9976539e85d1d70cd029929f9d
pi@raspberrypi:~$ docker ps
```

3) Enter “xhost +” in the terminal to start the accessing control of “X Server”.

```
pi@raspberrypi:~$ xhost +
access control disabled, clients can connect from any host
```

4) Enter “docker ps -a” in the terminal to check the ID of the newly created container.

```
pi@raspberrypi:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
76a092503f2e	ros:humble	"/ros_entrypoint.sh ..."	14 seconds ago	Up 13 seconds		humble
75b6634e6227	ros:melodic	"/ros_entrypoint.sh ..."	18 hours ago	Up 12 minutes		melodic
a24dbc912220	debian	"bash"	20 hours ago	Exited (0) 20 hours ago		blissful_mccar
hy						
f1b755532336	debian:latest	"/bin/bash"	20 hours ago	Exited (255) 19 hours ago		jolly_bassi
8313afaf6374	debian:latest	"/bin/bash"	20 hours ago	Exited (0) 20 hours ago		fervent_morse
d2d0e43b41eb	debian	"/bin/bash"	22 hours ago	Exited (255) 19 hours ago		crazy_bell
9cfc6a98a68c	debian:latest	"/bin/bash"	22 hours ago	Exited (255) 19 hours ago		jolly_edison

5) Enter “docker exec -it 76a092503f2e /bin/bash” in the terminal to access the container.

```
pi@raspberrypi:~$ docker exec -it 76a092503f2e /bin/bash
```

6) Enter “useradd -m -s /bin/bash ubuntu” in the terminal to create a new user.

```
root@raspberrypi:/# useradd -m -s /bin/bash ubuntu
```

7) Enter “passwd ubuntu” in the terminal to set a password for “ubuntu”. After successfully set, the new password needs to be input twice.

```
root@50e095a38fff:/# passwd ubuntu
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
```

8) Enter “usermod -aG sudo ubuntu” in the terminal. Add “sudo” to the new user “ubuntu” to give it superuser privileges.

```
root@raspberrypi:/# usermod -aG sudo ubuntu
```

9) Enter “sudo apt-get update -y && sudo apt-get upgrade -y” in the terminal to update the available software package list and upgrade the installed package on the system.

```
root@raspberrypi:/# sudo apt-get update -y && sudo apt-get upgrade -y
Get:1 http://snapshots.ros.org/melodic/final/ubuntu bionic InRelease [13.8 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports bionic InRelease [242 kB]
Get:3 http://snapshots.ros.org/melodic/final/ubuntu bionic/main arm64 Packages [1017 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports bionic-updates InRelease [88.7 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports bionic-backports InRelease [83.3 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports bionic-security InRelease [88.7 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports bionic/universe arm64 Packages [11.0 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports bionic/main arm64 Packages [1285 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports bionic/restricted arm64 Packages [572 B]
```

10) Enter “sudo apt-get install vim -y” in the terminal to install the “Vim” text editor.

```
root@raspberrypi:/# sudo apt-get install vim -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libgpm2 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
```

11) Enter “sudo apt-get install ros-humble-desktop-full -y” in the terminal to install the complete desktop environment of the the “ROS Humble” version.

```
root@raspberrypi:/# sudo apt-get install ros-melodic-desktop-full -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  autoconf automake autopoint autotools-dev avahi-daemon bind9-host blt bsdmainutils curl cython dbus dh-autoreconf dh-strip-nondeterminism dmsetup file fltk1.3-doc fluid fonts-lato fonts-liberation font
```

2.2 ROS2 Environment Test

1) Enter the “docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash” command to access the container. (Note: “55ce” is the ID of the container with the ROS2 environment.)


```
ubuntu@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash  
ubuntu@raspberrypi:~$
```

2) Enter the “source /opt/ros/humble/setup.bash” command to manually configure the ROS2 environment.

```
ubuntu@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash  
ubuntu@raspberrypi:~$ source /opt/ros/humble/setup.bash
```

3) Every time you open the terminal, execute step 2 above to load the working space. Enter the “echo “source /opt/ros/humble/setup.bash” >> ~/.bashrc” command and write it into the “.bashrc” file.

```
ubuntu@raspberrypi:~$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
```

4) Then enter the “source ~/.bashrc” command to take the “.bashrc” file into effect. This action frees you from loading the working space environment every time opening the terminal.

```
ubuntu@raspberrypi:~$ source ~/.bashrc  
ubuntu@raspberrypi:~$
```

5) Enter “roslaunch turtlesim turtlesim_node” to start the turtle GUI interface. Its successful starting indicates the successful installation of ROS2.

```
ubuntu@raspberrypi:~$ ros2 run turtlesim turtlesim_node
```

